

Vector Spaces and Subspaces

Historical Note

Giuseppe Peano wrote down the axioms of a vector space in 1888, in a slim book whose title announces its debt: *Calcolo geometrico secondo l'Ausdehnungslehre di H. Grassmann*. In its ninth chapter he defines what he calls a *sistema lineare*, lists essentially the eight rules of Definition 5.1 below, and then, with no fuss at all, observes that the functions on an interval form one. That is the modern definition, infinite-dimensional examples included, arrived at forty years before anyone else. It was almost entirely ignored. The idea had to be found again, by Hermann Weyl in *Raum, Zeit, Materie* in 1918 and by Stefan Banach in his thesis of 1922, before it entered the standard curriculum through van der Waerden's *Moderne Algebra* in 1930. Peano's book was building on Grassmann, whose own work had already waited thirty years for readers. Twice in a row, the abstraction arrived before the audience.

Everything so far has been about $\mathbb{R}^n$. But Example 2.4 already noticed that images, polynomials and word embeddings obey the same arithmetic, and Chapter 4 added matrices to the list. Rather than prove each theorem four times, we prove it once, for any object that adds and scales. This chapter says what that means, and then finds the subspaces hiding inside every matrix we have written down.

5.1 The axioms

Definition 5.1 (Vector space). A real vector space is a set V together with an addition $V \times V \rightarrow V$ and a scalar multiplication $\mathbb{R} \times V \rightarrow V$ such that, for all $u, v, w \in V$ and all $\lambda, \mu \in \mathbb{R}$:

(V1) $u + v = v + u$;

(V2) $(u + v) + w = u + (v + w)$;

(V3) there is an element $\mathbf{0} \in V$ with $v + \mathbf{0} = v$ for every v ;

(V4) for each v there is an element $-v$ with $v + (-v) = \mathbf{0}$;

(V5) $\lambda(\mu v) = (\lambda\mu)v$;

(V6) $1v = v$;

(V7) $\lambda(u + v) = \lambda u + \lambda v$;

(V8) $(\lambda + \mu)v = \lambda v + \mu v$.

The elements of V are called *vectors* and those of $\mathbb{R}$ *scalars*.

Nothing in the list mentions arrows, coordinates or dimension. That is the point: the axioms record only what we actually used in Chapters 2 to 4, and any theorem proved from them alone is true of every object satisfying them.

Key Idea

Replacing $\mathbb{R}$ by $\mathbb{C}$, or by any field, changes nothing in the definition and nothing in the proofs of this chapter; we work over $\mathbb{R}$ because that is where the pictures live. From Chapter 9 onwards we will be forced into $\mathbb{C}$, because that is where eigenvalues live, and it will cost us nothing to move because the theory was never really about $\mathbb{R}$ in the first place.

The axioms are terse enough that some familiar facts have to be earned rather than assumed.

Proposition 5.2 (First consequences). *In any vector space V :*

- (a) *the zero vector is unique, and so is $-v$ for each v ;*
- (b) *$0v = \mathbf{0}$ for every $v \in V$, and $\lambda \mathbf{0} = \mathbf{0}$ for every $\lambda \in \mathbb{R}$;*
- (c) *$(-1)v = -v$;*
- (d) *if $\lambda v = \mathbf{0}$ then $\lambda = 0$ or $v = \mathbf{0}$.*

Proof. (a) If $\mathbf{0}$ and $\mathbf{0}'$ both satisfy (V3) then $\mathbf{0} = \mathbf{0} + \mathbf{0}' = \mathbf{0}'$, using (V1) for the second step. If w and w' are both negatives of v , then $w = w + \mathbf{0} = w + (v + w') = (w + v) + w' = \mathbf{0} + w' = w'$.

(b) By (V8), $0v = (0 + 0)v = 0v + 0v$. Adding $-(0v)$ to both sides gives $\mathbf{0} = 0v$. The second statement is the same argument with (V7): $\lambda \mathbf{0} = \lambda(\mathbf{0} + \mathbf{0}) = \lambda \mathbf{0} + \lambda \mathbf{0}$.

(c) $v + (-1)v = 1v + (-1)v = (1 + (-1))v = 0v = \mathbf{0}$ by (V6), (V8) and part (b). So $(-1)v$ is a negative of v , and by (a) it is *the* negative.

(d) Suppose $\lambda v = \mathbf{0}$ and $\lambda \neq 0$. Then $v = 1v = (\lambda^{-1}\lambda)v = \lambda^{-1}(\lambda v) = \lambda^{-1}\mathbf{0} = \mathbf{0}$, using (V6), (V5) and part (b). $\square$

Caution

Part (d) is the only place where we divide, and it needs λ^{-1} to exist. Over a field it always does, which is why the theory is stated for fields. Over the integers, where 2 has no inverse, the analogous structure is called a module and part (d) genuinely fails: in the group $\mathbb{Z}/2\mathbb{Z}$ we have $2 \cdot 1 = 0$ with neither factor zero. Every later theorem that quietly divides by a scalar is relying on this.

5.2 A gallery

Example 5.3 (Spaces). *Each of the following is a real vector space under the obvious operations, and in each case the eight axioms hold because they hold for real numbers, one coordinate or one point at a time.*

- (i) $\mathbb{R}^n$, with Definition 2.2. The prototype.
- (ii) $M_{m \times n}(\mathbb{R})$, with Definition 4.1. Here $\mathbf{0}$ is the zero matrix.
- (iii) $\mathbb{R}[t]$, all polynomials with real coefficients, and P_n , those of degree at most n .
- (iv) All functions $f : S \rightarrow \mathbb{R}$ from any set S , with $(f + g)(s) = f(s) + g(s)$ and $(\lambda f)(s) = \lambda f(s)$. Taking $S = \{1, \dots, n\}$ recovers $\mathbb{R}^n$, so this example contains the prototype.
- (v) The continuous functions on $[0, 1]$, and separately the differentiable ones. Both sit inside (iv).
- (vi) The solutions of the differential equation $y'' + y = 0$. If y_1 and y_2 solve it so does $\lambda y_1 + \mu y_2$, because differentiation is linear.
- (vii) $\mathbb{C}$, regarded as a real vector space: complex numbers add, and may be scaled by reals.
- (viii) $\{0\}$, the space with one element. Small, but it satisfies every axiom.

Insight

Example (vi) explains a fact usually presented as folklore in a differential equations course: that the general solution of a second-order linear homogeneous equation is a combination of two independent solutions. It is not folklore. The solution set is a vector space, and Chapter 6 will prove that this particular one has dimension two, so *every* solution is $\lambda \cos t + \mu \sin t$ and there is nothing else to look for. Whenever a subject announces that its general solution is a linear combination of finitely many special ones, a finite-dimensional vector space is the reason.

Example 5.4 (Things that are not vector spaces). (i) $\mathbb{R}^2$ with the multiplication-like addition $(x, y) \oplus (x', y') = (xx', yy')$. The pair $(1, 1)$ does play the role of $\mathbf{0}$ in (V3), but (V4) fails: nothing can be added to $(0, 0)$ to recover $(1, 1)$, so $(0, 0)$ has no negative.

(ii) The first quadrant $\{(x, y) : x \geq 0, y \geq 0\}$. Closed under addition and under positive scaling, but $(-1)(1, 1)$ escapes it, so (V4) fails.

(iii) Polynomials of degree exactly 3. The zero polynomial is missing, and $(t^3 + 1) + (-t^3)$ has degree 0.

(iv) Vectors in $\mathbb{R}^2$ with integer entries. Adding is fine; scaling by $\frac{1}{2}$ is not.

(v) The invertible $n \times n$ matrices. Closed under multiplication, which is a different structure entirely, but $\mathbf{1} + (-\mathbf{1}) = \mathbf{0}$ is not invertible.

5.3 Subspaces

Most vector spaces we care about arrive as pieces of a larger one, and there is a short test for recognising them.

Definition 5.5 (Subspace). A subset $W \subseteq V$ is a subspace if it is a vector space under the operations inherited from V .

Proposition 5.6 (The subspace test). A subset $W \subseteq V$ is a subspace if and only if

(a) $\mathbf{0} \in W$;

(b) $u, v \in W \implies u + v \in W$;

(c) $\lambda \in \mathbb{R}$ and $v \in W \implies \lambda v \in W$.

Equivalently: W is nonempty and closed under taking linear combinations, $\lambda u + \mu v \in W$ for all $u, v \in W$ and all $\lambda, \mu \in \mathbb{R}$.

Proof. The conditions are necessary, since a subspace is a vector space and (b), (c) merely say the operations do not leave W . Conversely, suppose they hold. Axioms (V1), (V2), (V5) to (V8) are identities that hold throughout V , so they hold in particular for elements of W ; they are inherited for free. Only (V3) and (V4) could fail, and (a) supplies the zero while (c) with $\lambda = -1$ supplies negatives, by Proposition 5.2(c).

For the reformulation: closure under linear combinations gives (b) with $\lambda = \mu = 1$ and (c) with $\mu = 0$, and if W is nonempty, picking any $v \in W$ gives $\mathbf{0} = 0v \in W$, which is (a). $\square$

Key Idea

Check the zero first. It costs one glance, it is the condition that fails most often, and it rules out every set defined by an equation with a nonzero right-hand side. A subspace of $\mathbb{R}^n$ is a flat sheet through the origin; move it off the origin and it stops being one, which is exactly the distinction Theorem 3.14 drew between $Ax = 0$ and $Ax = b$.

Figure 5.1 slides one line off the origin and watches the test fail twice over: the zero vector is no longer in the set, and the sum of two of its points lands above it.

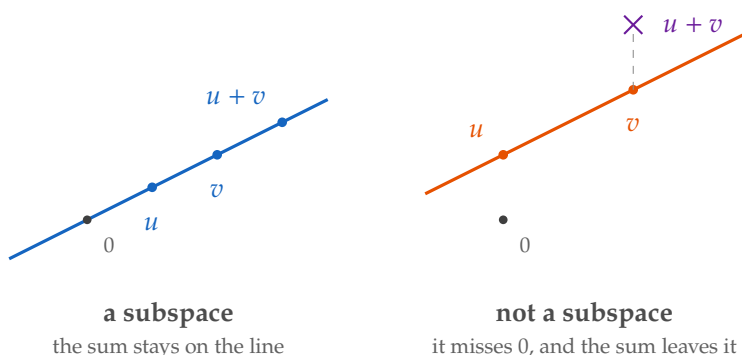


Figure 5.1. The same line, slid off the origin. On the left the line passes through $\mathbf{0}$ and is closed under addition and scaling. On the right it is the set of points with $y = \frac{1}{2}x + 1$; it fails the test twice over, since $\mathbf{0}$ is not on it and the sum of the two marked points lands above it.

Example 5.7 (Subspaces, and impostors). Inside $\mathbb{R}^3$: the plane $\{(x, y, z) : x + y + z = 0\}$ is a subspace, and the parallel plane $x + y + z = 1$ is not, because it misses the origin. Inside $M_n(\mathbb{R})$: the symmetric matrices form a subspace, since $(\lambda A + \mu B)^t = \lambda A^t + \mu B^t$; so do the upper triangular matrices, and so do the matrices of trace zero, since the trace is linear. The invertible matrices do not, by Example 5.4(v).

A subtler impostor: $\{A \in M_2(\mathbb{R}) : \det A = 0\}$ is closed under scalar multiplication, since $\det(\lambda A) = \lambda^2 \det A$. It is not closed under addition: both $\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$ and $\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$ have determinant 0, and their sum is the identity, whose determinant is 1. Closure under scaling alone is never enough.

5.4 Building new subspaces from old

Proposition 5.8 (Intersections and spans). *Let V be a vector space.*

- (a) *The intersection of any collection of subspaces of V is a subspace.*
- (b) *For any subset $S \subseteq V$, the set $\text{span } S$ of all finite linear combinations of elements of S is a subspace, and it is the smallest subspace containing S : it is contained in every subspace that contains S .*

Proof. (a) Let $\{W_i\}$ be subspaces and $W = \bigcap_i W_i$. Every W_i contains $\mathbf{0}$, so W does. If $u, v \in W$ and $\lambda, \mu \in \mathbb{R}$ then u, v lie in each W_i , hence $\lambda u + \mu v$ lies in each W_i , hence in W . Apply Proposition 5.6.

(b) A linear combination of linear combinations of elements of S is again one, after collecting terms, so $\text{span } S$ passes the test; it contains $\mathbf{0}$ as the empty combination, or as $0s$ for any $s \in S$. If W is a subspace with $S \subseteq W$, then W is closed under linear combinations, so it contains every element of $\text{span } S$. Hence $\text{span } S$ is contained in the intersection of all such W , and it is itself one of them, so it is the smallest. $\square$

Definition 5.9 (Sum of subspaces). *For subspaces $U, W \subseteq V$, their sum is $U + W = \{u + w : u \in U, w \in W\}$.*

Proposition 5.10. *$U + W$ is a subspace, and it is the smallest subspace containing both U and W ; equivalently $U + W = \text{span}(U \cup W)$.*

Proof. It contains $\mathbf{0} = \mathbf{0} + \mathbf{0}$, and $\lambda(u + w) + \mu(u' + w') = (\lambda u + \mu u') + (\lambda w + \mu w')$ is again a sum of an element of U and an element of W . Any subspace containing U and W must contain every $u + w$, so it contains $U + W$; and taking $w = \mathbf{0}$ shows $U \subseteq U + W$, likewise for W . The identification with $\text{span}(U \cup W)$ then follows from Proposition 5.8(b), both being the smallest subspace containing $U \cup W$. $\square$

Caution

The *union* of two subspaces is almost never a subspace. Take the two axes in $\mathbb{R}^2$: both are subspaces, $(1, 0)$ and $(0, 1)$ lie in the union, and $(1, 1)$ does not. In fact $U \cup W$ is a subspace precisely when one of the two contains the other, which Exercise 5.17 asks you to prove. This is why $U + W$, and not $U \cup W$, is the right way to combine subspaces, and it is the first sign that the lattice of subspaces behaves unlike a collection of ordinary sets.

Figure 5.2 shows what goes wrong if the union is used instead of the sum, and Figure 5.3 draws the smallest case worth drawing, two planes through the origin of $\mathbb{R}^3$ meeting along a line.

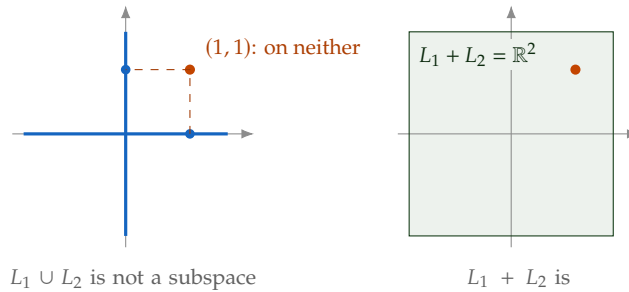


Figure 5.2. Why the definition asks for closure under addition, and why "sum" is not "union". The two axes are each a subspace, but their union is not: $(1, 0)$ and $(0, 1)$ both lie in it and $(1, 1)$ does not, so a single addition escapes the set. Repairing this is what Definition 5.9 does: $L_1 + L_2$ is everything reachable by adding one vector from each, which here is the whole plane. The intersection needs no repair, being $\{0\}$ and already a subspace, and that is general: intersections of subspaces are always subspaces and unions almost never are. The asymmetry is worth remembering, because it is why lattices of subspaces use $+$ and $\cap$ rather than $\cup$ and $\cap$.

5.5 The subspaces attached to a matrix

Every matrix carries three subspaces, and between them they encode everything Chapter 3 discovered about solving $Ax = b$.

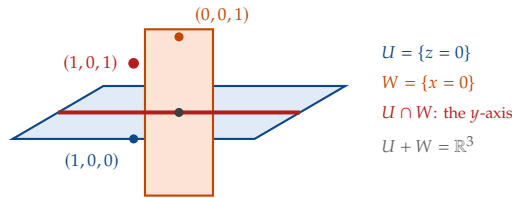
Definition 5.11 (Null space, column space, row space). For $A \in M_{m \times n}(\mathbb{R})$, define

$$N(A) = \{x \in \mathbb{R}^n : Ax = 0\}, \quad C(A) = \{Ax : x \in \mathbb{R}^n\} \subseteq \mathbb{R}^m,$$

and let the row space $R(A) \subseteq \mathbb{R}^n$ be the span of the rows of A . The set $N(A)$ is the null space or kernel, and $C(A)$ is the column space, image or range.

Proposition 5.12. $N(A)$ is a subspace of $\mathbb{R}^n$, and $C(A)$ is a subspace of $\mathbb{R}^m$ equal to the span of the columns of A . Moreover $Ax = b$ is consistent if and only if $b \in C(A)$.

Proof. $A0 = 0$, and $A(\lambda x + \mu y) = \lambda Ax + \mu Ay = 0$ whenever $Ax = Ay = 0$, so $N(A)$ passes the subspace test. For $C(A)$, the column view of Proposition 4.4 with x a single



$\dim U = 2$, $\dim W = 2$, $\dim(U \cap W) = 1$, $\dim(U + W) = 3$
 and $2 + 2 - 1 = 3$, which is the dimension formula this chapter is heading towards

the union is *not* a subspace: $(1, 0, 0) \in U$ and $(0, 0, 1) \in W$, but their sum $(1, 0, 1)$ has $z \neq 0$ and $x \neq 0$,
 so it lies in neither plane. It does lie in $U + W$, and that is the difference between $U \cup W$ and $\text{span}(U \cup W) = U + W$

two planes through the origin meet in a line and span everything, and the smallest subspace
 containing both is their sum, not their union

Figure 5.3. Definition 5.9 and Proposition 5.10 in the smallest case worth drawing: two planes through the origin of $\mathbb{R}^3$, the horizontal $U = \{z = 0\}$ and the vertical $W = \{x = 0\}$. They meet along the y -axis, drawn in coral, so the intersection is a line and is itself a subspace, as Proposition 5.8(a) requires. Their sum is all of $\mathbb{R}^3$, and the four dimensions satisfy $2 + 2 - 1 = 3$. The three marked points make the essential distinction. The point $(1, 0, 0)$ lies in U and the point $(0, 0, 1)$ lies in W , so both lie in the union; but their sum $(1, 0, 1)$ has a nonzero z coordinate and a nonzero x coordinate, so it lies in neither plane and therefore not in the union. A union of subspaces is almost never a subspace, which is exactly why Definition 5.9 defines the sum instead, and Proposition 5.10 identifies it as the span of the union: the smallest subspace that repairs the failure.

column says $Ax = x_1A_1 + \cdots + x_nA_n$, so the set of all Ax is exactly $\text{span}\{A_1, \dots, A_n\}$, a subspace by Proposition 5.8(b). The last claim is a restatement: $Ax = b$ has a solution exactly when b is one of the vectors Ax . $\square$

The two spaces do not live in the same place, and Figure 5.4 is worth a look before going on: the null space sits in the source $\mathbb{R}^n$ and records what A destroys, while the column space sits in the target $\mathbb{R}^m$ and records what A produces.

Proposition 5.13 (Elimination preserves the row space). *If B is obtained from A by elementary row operations, then $R(B) = R(A)$. Consequently the nonzero rows of any echelon form of A span $R(A)$.*

Proof. Each row of $B = EA$ is, by Proposition 4.4, a linear combination of the rows of A , so $R(B) \subseteq R(A)$. Since E is invertible, $A = E^{-1}B$ and the same argument gives $R(A) \subseteq R(B)$. $\square$

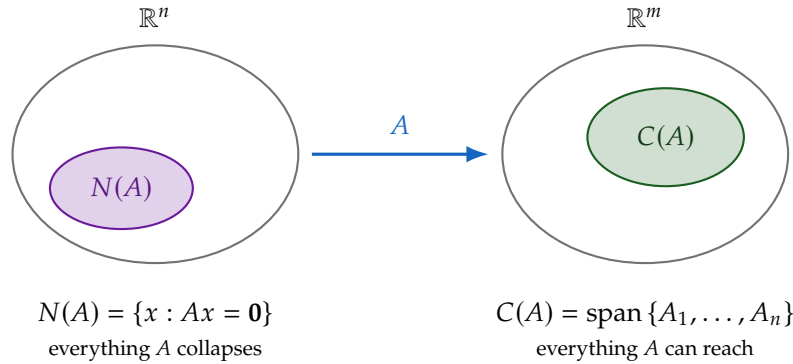


Figure 5.4. The two subspaces live in different places, and confusing them is the commonest error of the next few chapters. The null space sits in the source $\mathbb{R}^n$ and records what A destroys; the column space sits in the target $\mathbb{R}^m$ and records what A produces. Both contain the origin of their own space, as every subspace must. The row space, not drawn, sits alongside $N(A)$ in $\mathbb{R}^n$.

Caution

Row operations do *not* preserve the column space. With $A = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$ one step gives $B = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}$, and $C(A) = \text{span} \{(1, 1)\}$ while $C(B) = \text{span} \{(1, 0)\}$: different lines. What survives is not the column space but the *relations* among the columns, by Lemma 4.19. That is why the recipe below reads the pivot positions off the echelon form and then takes the corresponding columns of the *original* matrix.

Example 5.14 (All three subspaces of one matrix). *Let*

$$A = \begin{pmatrix} 1 & 2 & 0 & 1 \\ 2 & 4 & 1 & 4 \\ 3 & 6 & 1 & 5 \end{pmatrix}.$$

Eliminate with $R_2 \rightarrow R_2 - 2R_1$, $R_3 \rightarrow R_3 - 3R_1$, then $R_3 \rightarrow R_3 - R_2$:

$$\rightarrow \begin{pmatrix} 1 & 2 & 0 & 1 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 1 & 2 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 2 & 0 & 1 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

Pivots sit in columns 1 and 3.

Row space. By Proposition 5.13 the two nonzero rows of the echelon form span it:
 $R(A) = \text{span} \{(1, 2, 0, 1), (0, 0, 1, 2)\} \subseteq \mathbb{R}^4$.

Null space. The free unknowns are x_2 and x_4 . The second row gives $x_3 = -2x_4$, and the first gives $x_1 = -2x_2 - x_4$, so

$$N(A) = \text{span}\{(-2, 1, 0, 0), (-1, 0, -2, 1)\} \subseteq \mathbb{R}^4.$$

Check the second generator against the original A : the rows give $-1 + 0 + 0 + 1 = 0$, $-2 + 0 - 2 + 4 = 0$ and $-3 + 0 - 2 + 5 = 0$.

Column space. Take the columns of the original A in the pivot positions: $C(A) = \text{span}\{(1, 2, 3), (0, 1, 1)\} \subseteq \mathbb{R}^3$. The other two columns are combinations of these, as they must be: column 2 is 2 times column 1, and column 4 is $(1, 4, 5) = 1 \cdot (1, 2, 3) + 2 \cdot (0, 1, 1)$. Notice the bookkeeping. Two pivots, and each of the three subspaces is described by exactly two generators, except the null space, which needs $4 - 2 = 2$ as well. That is not a coincidence, and naming it is the business of the next chapter.

Insight

A nontrivial null space is a statement about your data, not about your algebra. If A is a design matrix whose rows are observations and whose columns are features, then $x \in N(A)$ is a combination of features that every observation reports as zero: the data cannot see it. Two parameter vectors differing by such an x fit the data identically, so no amount of optimisation will choose between them. This is exactly Theorem 3.14, and Proposition 3.13 says it is guaranteed whenever there are more features than observations. Regularisation, in this language, is the act of adding a preference so that the solution set of one point is picked out of a whole flat.

5.6 What comes next

We now have the objects but not yet the measurements. Example 5.14 produced two generators for each subspace, and the obvious questions are whether two was necessary, whether some other elimination would have produced three, and what the number two is really counting. Answering them requires the notions of linear independence, basis and dimension, which are the subject of the next chapter, and they will show that the count is forced: it is the rank of Theorem 3.9, looked at from three different directions.

Open Problem 5.15 (The largest subset of a vector space with no line in it). Work in $\mathbb{F}_q^n$, a vector space over a finite field. Call a subset a *cap* if it contains no three points lying on a common line, that is, no three distinct x, y, z with $x + y + z = 0$ when $q = 3$. Write $r_q(n)$ for the largest cap. What is the growth rate of $r_q(n)$ as $n \rightarrow \infty$?

What is known. For $q = 3$ the problem is famous, and in 2016 the polynomial method of Croot, Lev and Pach, adapted by Ellenberg and Gijswijt, produced the first exponential improvement in decades: $r_3(n) \leq 2.756^n$ [220]. The best construction gives $r_3(n) \geq 2.217^n$. Those two numbers have not met, and for $q \geq 4$ the gap is wider still. The exact exponential growth rate is unknown for every q .

Why it lives in this chapter. The proof is pure linear algebra of the kind this chapter sets up, and it is startlingly short. Consider the vector space of polynomials on $\mathbb{F}_3^n$ of degree at most d in which no variable appears to a power above 2. Its dimension is a sum of multinomial coefficients, which one can count. Now show that a cap forces a certain matrix built from these polynomials to be diagonal, so its rank is the size of the cap, and the rank is also at most the dimension of the polynomial space. A bound on a combinatorial object came out of a dimension count on a space of functions, which is the entire method of this chapter used at full stretch.

Where a reader might start. Compute $r_3(n)$ by hand for $n = 1, 2$ and by computer for $n = 3, 4$, where the answers are 2, 4, 9, 20. Then read the Ellenberg and Gijswijt argument with Chapter 6's dimension counting in hand; the only tool it uses beyond this chapter is that the rank of a diagonal matrix is the number of nonzero diagonal entries.

Exercises

Exercise 5.16. Which of these are subspaces of $\mathbb{R}^3$? Justify each answer, and when a set fails, say which condition of Proposition 5.6 it violates.

- (a) $\{(x, y, z) : x + y + z = 0\}$
- (b) $\{(x, y, z) : x + y + z = 1\}$
- (c) $\{(x, y, z) : xyz = 0\}$
- (d) $\{(x, y, z) : x = y\}$
- (e) $\{(x, y, z) : x \geq 0\}$

Exercise 5.17. Let U and W be subspaces of V . Show that $U \cup W$ is a subspace if and only if $U \subseteq W$ or $W \subseteq U$. Hint: if neither containment holds, pick $u \in U \setminus W$ and $w \in W \setminus U$ and ask where $u + w$ can live.

Exercise 5.18. In $M_2(\mathbb{R})$, let S be the symmetric matrices and K the antisymmetric ones. Show that both are subspaces, that $S + K = M_2(\mathbb{R})$ and that $S \cap K = \{0\}$. Compare with

Exercise 4.29.

Exercise 5.19. For $A = \begin{pmatrix} 1 & 3 & 2 \\ 2 & 6 & 5 \\ 1 & 3 & 3 \end{pmatrix}$, find generators for $N(A)$, $C(A)$ and $R(A)$, and verify your null-space generator against the original matrix.

Exercise 5.20. Show that $\{A \in M_n(\mathbb{R}) : \text{trace } A = 0\}$ is a subspace. Example 5.7 showed that $\{A : \det A = 0\}$ is not a subspace of $M_2(\mathbb{R})$; extend that argument to every $n \geq 2$, and say which of the three conditions of Proposition 5.6 the set does satisfy.

Exercise 5.21. Show that the set of twice-differentiable functions satisfying $y'' + y = 0$ is a vector space, and verify that $\sin$ and $\cos$ belong to it. Is the set of solutions of $y'' + y = 1$ also a vector space? Relate your answer to Theorem 3.14.

Exercise 5.22. Give a 2×2 matrix A and a single row operation after which the column space is different. Then explain why no row operation can change $N(A)$ or $R(A)$.

Exercise 5.23. Prove Proposition 5.2(d) again, writing beside each step the number of the axiom it uses. Then find where your proof would break if the scalars were the integers rather than $\mathbb{R}$.

Exercise 5.24. Let $U = \text{span}\{(1, 0, 0), (0, 1, 0)\}$ and $W = \text{span}\{(0, 1, 0), (0, 0, 1)\}$ in $\mathbb{R}^3$. Describe $U \cap W$ and $U + W$ explicitly, and check that neither $U \cup W$ nor the set difference $U \setminus W$ is a subspace.

References and further reading

Peano's 1888 axioms, and the long silence that followed them, are recounted in most histories of the subject; the connection back to Grassmann in Chapter 2 is Peano's own, and is announced in his title. Our development is standard and written from scratch, but the order of presentation, axioms first and matrix subspaces immediately afterwards, is a deliberate choice: Hefferon [307] and Austin [35] both introduce subspaces of $\mathbb{R}^n$ long before the abstract definition, which is gentler, while Axler [37] begins with the axioms and never looks back, which is cleaner. We have tried to get the benefit of both by proving the general statement and then immediately spending it on $N(A)$ and

$C(A)$. Treil [610] is unusually explicit about why the abstract definition earns its keep. Banica [51], our baseline, moves through this material quickly on the way to operator theory, where the same axioms are carrying Hilbert spaces rather than $\mathbb{R}^n$; readers who want to see where the abstraction is eventually going will find that instructive.